

A Differential Neural Scene Codec: Formal Conditions and Exploratory Limits of Scene-Conditioned Residual Neural Control

Author: Micah Blumberg

Self Aware Networks Research Institute

<https://selfawarenetworks.com>

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Abstract

A bidirectional perceptual interface must solve a harder problem than neural decoding or stimulation alone. It must estimate a person's current measurable neural state, construct a target associated with a scene, determine which part of their difference is observable and reachable, and apply a bounded update. Micah Blumberg's April 11, 2017 Neural Lace Podcast proposed the core operation in plain language: do not place a complete artificial scene pattern on top of an incompatible ongoing pattern; estimate the difference between the current and intended patterns and transmit only that difference. The same source proposed Alternating Transmission Protocol (ATP), a deliberate dual reference to feedback-versus-broadcast communication and adenosine triphosphate. A 2022 Self Aware Networks source expanded this genealogy through participant-specific phase differences, multimodal object states, Phase Wave Differentials (PWD), and receive-transform-project neural rendering.

This paper turns that historical operation into a falsifiable control framework. It inherits state estimation, feedback, model-predictive control, visual decoding, and closed-loop stimulation, then defines the narrower conjunction that requires a scene-conditioned target, person-specific state estimate, observable-and-reachable residual, uncertainty governor, and explicit no-control option. It preserves a 3,000-run exploratory benchmark and a separately frozen 2,520-run adversarial benchmark. In the adversarial test, predictive residual control won two of six conditions, ranked fifth under observation-rank loss, lost to no control under combined stress, and crossed a state bound in 16.1 percent of its runs.

A new development-only public-data test evaluates two noninterventive components on 14 Allen Visual Coding Neuropixels sessions. A nearest-centroid reader classified 118 natural-scene frames with median held-block top-1 accuracy 0.343 (session-bootstrap 95 percent interval 0.260-0.413), versus 0.00847 uniform chance and 0.00840 after training-label permutation. A training-fold scene-to-response target model reduced mean-squared error by a median 15.6 percent relative to the global-mean comparator, with positive gain in all 14 sessions. This supports measurable scene information and a scene-conditioned target estimate in the development data. It does not test stimulation, residual writing, perception, behavior, consciousness, or independent replication. The complete result therefore narrows rather than completes the codec: reading and target estimation are empirically feasible components, while safe causal writing remains the decisive unsolved test.

Scope and evidence boundary

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1. The engineering problem

Visual reconstruction studies have shown that selected features of viewed images and natural movies can be inferred from fMRI activity [3-5]. Closed-loop stimulation studies have shown that measured brain activity can trigger or tune subsequent stimulation [6,7]. These are important components of a bidirectional interface, but their coexistence does not by itself define a perceptual codec.

A codec in the present sense requires four linked transformations:

1. map an external scene or intended percept into a target neural representation;
2. estimate the user's current neural representation from incomplete and noisy measurements;
3. compute an update that depends on both target and current state; and
4. write only an admissible component of that update through a bounded control channel.

The term *scene* is intentionally broader than retinal pixels. A scene may include color, shape, texture, taste, smell, touch, body state, and anticipated action. The term *neural state* is intentionally narrower than consciousness. It denotes only the modeled variables available to a specified measurement and intervention system. No equation in this paper turns an unobserved total brain state into a measurable object by definition.

The central hypothesis is conditional:

For a defined scene family, participant, measurement system, and intervention channel, a controller that writes an uncertainty-bounded, scene-conditioned residual may improve a preregistered task objective relative to matched decoder-only, open-loop, feedforward, and conventional feedback controls.

That hypothesis can fail for conceptual reasons before any experiment begins. The target may not be identifiable. The reachable neural subspace may not carry the relevant perceptual information. Stimulation artifacts may masquerade as state correction. A lower-complexity controller may achieve the same behavior. Those possibilities are part of the model, not qualifications added after a negative result.

2. Source genealogy

2.1 The 2017 network-protocol question

The April 11, 2017 Neural Lace Podcast asked whether communication in the brain was more like TCP, with connection and feedback, or UDP, with faster connectionless transmission and tolerance for dropped packets [1]. These were engineering analogies, not claims that neurons literally implement Internet standards. The proposed answer was that biological communication may alternate between feedback-dependent and broadcast-like modes.

The source named this proposed architecture **Alternating Transmission Protocol**. The acronym ATP was deliberate: the transcript immediately noted that adenosine triphosphate is the cell's energy currency [1]. The name therefore joined two questions that remain separable experimentally: how neural information is routed and how communication expenditure is metabolically supported. This paper preserves that double meaning as intellectual genealogy; it does not use ATP concentration as a control variable.

2.2 The 2017 differential scene operation

Later in the same episode, the public audio considered adding an artificial steak percept. It warned against sending a complete target pattern on top of an ongoing, potentially incompatible pattern. The proposed operation was to estimate how the current pattern differs from the pattern appropriate to the intended scene and transmit that difference [1]. In modern notation, this is the ancestor of a scene-conditioned residual update. This paper paraphrases the audio because legacy machine transcripts contain material errors; they are discovery aids, not quotation authority.

The source is important because it joins three operations in one dated public record: scene modeling, current-state estimation, and difference transmission. It does not supply a solved inverse model, a stimulation map, a safety proof, or an experimental result. Those missing elements define the work of this paper.

2.3 The 2022 PWD and NAPOT extension

The August 2022 note `a0306z.md` developed the same problem through a multimodal broccoli example [2]. A recognizable object was decomposed into color, shape, texture, feel, crunch, smell, and taste. The note proposed that two people can recognize the same object despite differing detailed neural encodings, and asked whether an interface could intercept, translate, and recreate the relevant distributed differences. It connected those differences to Phase Wave Differentials (PWD) and to a receive-transform-project account later organized as Neural Array Projection Oscillation Tomography (NAPOT).

For this paper, PWD supplies a candidate feature family whose distinctive role is receiver-relative change, not a synonym for phase coding or traveling waves. NAPOT supplies the proposed receive-transform-project architecture that links partial measurements, recurrent transformation, and distributed updates. The codec framework deliberately permits those mechanisms to fail: another measurable representation can instantiate the engineering test.

2.4 Provenance and transcript governance

The April 2017 publication date is anchored by the public episode [1]. The original recording controls what was said. A newer, clearer machine retranscription may assist semantic reading, but every load-bearing quotation must be checked against that recording. Older transcripts and captions are used only for discovery and version history because serious transcription errors are known to exist. The August 2022 extension is fixed by an immutable Git commit [2]. Neither later terminology nor later transcription is backdated into the earlier source.

3. Relation to established work

Miyawaki et al. reconstructed visual patterns by combining multiscale local decoders [3]. Nishimoto et al. inverted an encoding model to reconstruct coarse natural-movie content from slow fMRI responses [4]. Kay et al. identified viewed natural images with Bayesian reconstruction [5]. These studies establish that measurement-dependent stimulus information can sometimes be decoded. They do not establish a complete or person-independent neural target for a percept.

Berenyi et al. used detected epileptic activity to trigger transcranial electrical stimulation in rats [6]. Soleimani et al. reviewed and framed closed-loop transcranial electrical stimulation with concurrent fMRI as a route toward precision neuromodulation [7]. These studies and reviews establish that brain measurement, error or state criteria, and stimulation can be joined in a feedback system. Generic closed-loop control is therefore prior art.

State estimation, feedback, optimal control, and model-predictive control are also established mathematical fields [8,9]. The equations below inherit that machinery. The narrower contribution tested here is the conjunction of:

- a scene-conditioned target rather than a symptom threshold alone;
- a person-specific current-state estimate;
- a residual defined only in an observable and reachable projection;
- explicit uncertainty suppression and intervention bounds; and
- a codec interpretation that connects a 2017 perceptual read-model-write proposal to a reproducible falsifier program.

If that conjunction offers no advantage over a simpler feedforward or standard feedback controller, the distinctive engineering claim is not supported.

4. Formal model

4.1 Scene, latent state, and measurement

Let $x_t \in \mathcal{X}$ denote an external scene or specified intended perceptual condition at time t . Let $z_t \in \mathbb{R}^n$ denote the participant-specific latent state used by the model. Measurements are

$$y_t = H_i(z_t) + \eta_t. \quad [1]$$

where H_i is the measurement operator for participant (i), and η_t includes sensor noise, physiological variability, and unresolved artifact. Equation (1) does not imply that H_i captures the total neural state.

An estimator uses the measurement history, the scene history, and previously applied controls:

$$(\hat{z}_t, \Sigma_t) = E_i(y_{0:t}, x_{0:t}, u_{0:t-1}). \quad [2]$$

where $\hat{z}_t$ is an estimated state and Σ_t is an uncertainty descriptor. A decoder that reports only a point estimate is insufficient for the uncertainty-bounded controller proposed here.

4.2 Scene-conditioned target

A participant-specific target model maps the scene to an equivalence class of neural states:

$$\mathcal{T}_{i,t} = D_i(x_t, c_t). \quad [3]$$

where c_t includes task, body, and behavioral context. The target is a set, not necessarily one exact vector, because multiple neural configurations may support behaviorally equivalent scene representations.

Let PO project onto the state components observable at the available measurement scale, and let PR project onto components reachable through the specified intervention channel. The codec may act only on their intersection:

$$P = PRPO, \quad r_t = P(z_{i,t} - \hat{z}_t), \quad z_{i,t} \in \mathcal{T}_{i,t}. \quad [4]$$

Equation (4) is the operational residual. It is not the difference between two complete experiences; it is the difference between modeled projections.

4.3 Dynamics and control

The latent dynamics are

$$z_{t+1} = F_i(z_t, u_{t-d}, w_t). \quad [5]$$

where d is intervention delay and w_t is process disturbance. A bounded controller proposes

$$\tilde{u}_t = \pi_i(r_t, \hat{z}_t, \Sigma_t, d), \quad u_t = \mathcal{G}(\tilde{u}_t, \hat{z}_t, \Sigma_t). \quad [6]$$

where $\mathcal{G}$ is a governor that can attenuate, clip, or cancel the command. The governor has authority over the tracking objective.

For the synthetic predictive-residual controller, the uncontrolled state is projected across the delay horizon:

$$\hat{z}_{(0)t+d+1} = I_{d+1}\hat{z}_t, \quad \tilde{u}_t = g(\Sigma_t, d)\hat{B}_i^+(z_{i,t}^* - \hat{z}_{(0)t+d+1}). \quad [7]$$

where $\hat{B}_i^+$ is a pseudoinverse and g is an uncertainty- and delay-dependent gain. This is one implementation, not the definition of the codec.

4.4 Objective and stopping rule

A controller is evaluated with a preregistered composite objective:

$$J = \sum_t [\|P(z_t - z_{i,t})\|_Q^2 + \lambda_u \|u_t\|_2^2 + \lambda_s 1\{z_t \notin \mathcal{S}_i\}]. \quad [8]$$

where $\mathcal{S}_i$ is a prospectively defined admissible state set. The controller must stop when uncertainty or a predicted boundary crossing exceeds a frozen threshold:

$$u_t = 0 \text{ if } U(\Sigma_t) > \tau_U \text{ or } \Pr(z_{t+1} \notin \mathcal{S}_i \mid y_{0:t}) > \tau_S. \quad [9]$$

The synthetic implementation uses simpler norm thresholds; it does not estimate a calibrated biological probability.

5. Preconditions: what must be true

5.1 Observability

For a local linearization $z_{t+1} = A z_t + B u_t$, $y_t = C z_t + \eta_t$, full-state observability requires

$$\mathcal{O}_n = [C^T \ (CA)^T \ \dots \ (CA^{n-1})^T]^T, \quad \text{rank}(\mathcal{O}_n) = n. \quad [10]$$

If this condition fails, the paper permits control only of an identifiable projection. A decoder score on held-out scenes is not proof that the causal state required for writing has been identified.

5.2 Reachability

For the same local model, full reachability requires

$$\mathcal{R}_n = [B \ AB \ \dots \ A^{n-1}B], \quad \text{rank}(\mathcal{R}_n) = n. \quad [11]$$

If the target residual lies outside the reachable subspace, clipping it is not a solution. The target must be projected, rejected, or redefined.

5.3 Identifiability and transport across people

The broccoli problem exposes an important ambiguity: two people may report and act on the same object while using neural states related by an unknown transformation. Let M_{ij} be a learned alignment between participant spaces. A cross-person target is admissible only if it preserves preregistered task relations on held-out scenes:

$$\mathcal{L}_{\text{heldout}}(D_j(x), M_{ij}D_i(x)) \leq \varepsilon M. \quad [12]$$

Failure of Equation (12) blocks cross-person writing. It does not justify assuming a universal code.

6. Synthetic benchmark program

6.1 Design

The benchmark is a deterministic linear-state sanity test, not a neural or clinical simulation. It uses six latent dimensions, eight measurements, 360 time steps, eight scene blocks, 100 seeds, five conditions, and six controllers. The 3,000 runs compare:

- decoder only, with no control;
- baseline feedback, without scene targeting;

- scene feedforward control;
- proportional residual control;
- uncertainty-gated model residual control; and
- delay-aware predictive residual control.

The conditions are nominal operation, increased measurement noise, subject / model shift, a two-step intervention delay, and combined noise-shift-delay stress. Commands are component- and norm-bounded. The simulator records tracking RMSE, mean control energy, clamp fraction, state-bound violations, stop fraction, and settling time.

The source code and complete per-run results are provided with the paper [12]. All metrics were finite and all 3,000 expected rows were produced.

6.2 Adaptation disclosure

The current configuration is Version 2. Version 1 exposed state-bound crossings in four of 500 predictive-controller runs. After seeing that result, the predictive stop threshold, prospective state check, delayed-command clearing, delay gain, and uncertainty gate were changed. Both generations are retained [12]. Version 2 must therefore be interpreted as exploratory controller design, not confirmatory evaluation of a frozen protocol.

6.3 Results

Table 1 reports median tracking RMSE and mean control energy for the two most informative controllers.

Condition	Predictive residual RMSE	Feedforward RMSE	Predictive energy	Feedforward energy	Predictive wins RMSE?
Nominal	0.1183	0.3097	0.2733	0.1270	Yes
Measurement noise	0.4272	0.3128	0.9170	0.1266	No
Subject shift	0.1831	0.4251	0.2772	0.1369	Yes
Two-step delay	0.3362	0.4133	0.1753	0.1265	Yes
Combined stress	0.4089	0.5065	0.1517	0.1347	Yes

Under two-step delay, the naive model-residual controller reached median RMSE 1.9056 and energy 4.6624, while the predictive residual reached 0.3362 and 0.1753. This shows that residual feedback without delay treatment can be substantially worse than feedforward control in the selected model.

Measurement noise produced the clearest adverse result. Predictive residual control had worse tracking and over seven times the median energy of scene feedforward control. A controller that repeatedly reacts to noisy residuals can amplify estimation error. Uncertainty gating in the current implementation did not solve that problem.

Median state-violation fraction for the predictive controller was zero in every condition, but 0.4 percent of its 500 runs contained at least one state-bound crossing. The correct conclusion is that the exploratory governor reduced but did not eliminate violations.

6.4 Frozen Draft 2 adversarial design

Draft 2 freezes a new configuration before generating results. The config hash is E538C16AC486FD87E2D6438CFB95993BFAFA199EB41F2A996F472B3A15AC841A. Sixty new seeds, six adversarial conditions, seven controllers, and 420 steps per run produce 2,520 trials. The six Draft 1 controllers remain, and a regularized one-step model-predictive controller is added. The conditions are:

1. nonlinear latent dynamics;

2. a two-dimensional loss of observation rank;
3. a target component with near-zero control authority;
4. temporally correlated measurement noise;
5. stimulation artifact added to the measured signal; and
6. a combined condition containing every stressor plus a two-step delay.

Controller superiority is not a mechanical pass gate. The frozen software gate requires only complete declared coverage, finite metrics, and no biological or participant data. The runner produced all 2,520 rows. Its result JSON hash is FACCF2925966D13684B4156508356038CEE358A86C7CA54D2C0B8F83CDC012AC.

6.5 Frozen adversarial results

Condition	Lowest median-RMSE controller	Winning RMSE	Predictive RMSE	Predictive rank of 7	Predictive runs with a crossing
nonlinear dynamics	predictive residual	0.1962	0.1962	1	0/60
observation rank loss	scene feedforward	0.2792	1.4428	5	33/60
unreachable target	ridge MPC	1.5619	1.7195	4	0/60
correlated noise	scene feedforward	0.3301	0.3957	2	0/60
stimulation artifact	predictive residual	0.2993	0.2993	1	0/60
combined adversarial	decoder only	1.8061	2.6138	4	23/60

The combined condition is the strongest negative result: applying no control had lower median tracking error than every active controller. Predictive residual control also failed sharply under rank loss. Its median tracking RMSE was 1.4428 versus 0.2792 for scene feedforward, and 55 percent of its runs crossed the state bound at least once. Across all 360 predictive-controller runs, 58 had a crossing, or 16.1 percent. A zero median violation rate in four conditions does not neutralize those failures.

The unreachable-target condition distinguishes total error from error on the nominally reachable coordinates. Ridge MPC had total RMSE 1.5619 and reachable coordinate RMSE 0.2040; predictive residual control had 1.7195 and 0.9594. The target’s unreachable component therefore cannot be treated as a control failure that more gain should solve. It must be projected out, rejected, or used to invalidate the target contract.

7. Public visual-data component test

7.1 The bounded question

Before asking whether a system can write a scene, two simpler questions must be answered. First, does the measured population contain held-out information about which scene was shown? Second, can a scene label predict a participant- and session-specific target response better than a scene-independent average? These are the reader and target-model components of the codec. A positive result cannot establish the stimulation or residual-writing component.

7.2 Dataset and frozen development protocol

The test used the Allen Institute Visual Coding Neuropixels dataset described by Siegle et al. [10] and preserved as DANDIset 000021, immutable version 0.251116.2246 [11]. Fourteen development sessions were selected before outcome inspection. Five separately designated final-test sessions remained sealed. Only primary visual cortex units, nonblank natural-scene presentations, and 118 frame identities were used.

For each presentation, spike counts were aggregated from the governed 20-ms-bin development artifacts. The three complete natural-scene blocks defined three folds: train on two blocks and test on the third. The

reader was a training-fold-z-scored nearest-class-centroid model. Its controls were uniform chance and a training-label permutation. The target model was the training-fold mean VISp response vector for each frame. Its controls were the global training mean and permuted frame-identity means. Uncertainty intervals are descriptive 95 percent percentile intervals from 10,000 session-level bootstrap draws, not confirmatory population inference.

7.3 Results

Component	Development result across 14 sessions	Comparator
Read top-1 accuracy	0.343 [0.260, 0.413]	chance 0.00847; shuffled 0.00840
Read top-5 accuracy	0.604 [0.490, 0.669]	chance top-5 0.0424
Target MSE gain	0.156 [0.122, 0.159]	global training mean
Target MSE gain	0.177 [0.150, 0.189]	permuted frame means
Target Poisson-deviance gain	0.115 [0.093, 0.124]	global training mean

The reader exceeded its controls in every development session. The target model also produced positive MSE and Poisson-deviance gains in every session. All 42 held-block folds completed. The result file, fold table, immutable stimulus metadata hashes, configuration, and deterministic runner are packaged with the paper [12]. Normal and optimized Python executions reproduced the same canonical result hash: 87017DABDE85A5EE9702076A3C5D35F36B01A29BE34860EFD73DC2F4B1D6CEF1.

7.4 What the result establishes and leaves open

The test establishes a bounded component result: these mouse VISp measurements carry held-block natural-scene information, and a scene-conditioned response target outperforms scene-independent and label-permuted comparators in the development sessions. It does not establish that the decoded state is a conscious percept, that it transports across animals or humans, or that an intervention can safely move the neural state toward the target. The experiment contains no stimulation, behavior, report, or consciousness endpoint. It is a development result, not an independent replication, and the sealed final sessions remain unopened.

8. Interpretation

The two benchmark generations support implementation feasibility only at the level of small synthetic models. Draft 1 showed that delay awareness matters and that residual feedback can lose under measurement noise. Draft 2 adds the more important result that residual control can become unsafe or counterproductive when the relevant state is not observable, the target is not reachable, or several model failures occur together. The correct controller can be feedforward, regularized MPC, predictive residual, or no control, depending on the declared failure regime.

None of these findings validates PWD, NAPOT, perceptual writing, or a biological ATP protocol. In Draft 1, the target state is supplied directly, the observation matrix is favorable, the intervention matrix is invertible enough for a pseudoinverse, and the dynamics are linear and stationary. Draft 2 deliberately violates several of those conveniences and the headline advantage largely disappears. Real neural systems add harder nonstationarity, unknown artifacts, partial causal access, biological safety constraints, and uncertain perceptual targets.

The most important implication is methodological. A serious scene codec cannot be evaluated by showing only an evocative reconstruction or a shift in a neural metric. It must demonstrate that:

- the scene target generalizes to held-out stimuli;
- the estimated residual is reliable under artifact and drift;
- the intervention affects the modeled state rather than the sensor;
- behavior or perception changes in a preregistered direction;
- the effect exceeds simpler controllers at matched energy and risk; and
- the result survives participant shift or remains explicitly person-specific.

9. Falsifiers and decisive tests

The distinctive hypothesis is rejected or narrowed if any of the following occurs under a frozen protocol:

1. **No observable residual.** The target-current difference cannot be decoded above a leakage-controlled null on held-out scenes.
2. **No reachable residual.** The intervention channel cannot move the observable state toward the target more than sham or matched random control.
3. **No residual advantage.** Feedforward, conventional feedback, or a simpler non-neural controller matches performance, energy, and violations.
4. **Artifact explanation.** Blanking, hardware, motion, expectancy, or stimulation artifact explains the apparent state correction.
5. **Delay failure.** Performance collapses under realistic sensing, computation, and actuation delay.
6. **Participant transport failure.** Cross-person alignment fails on held-out scene relations, blocking a shared code.
7. **Percept-behavior dissociation.** Neural tracking improves without the preregistered perceptual or behavioral outcome.
8. **Unsafe uncertainty response.** State-bound crossings or energy costs exceed the comparator despite the governor.

A result can falsify the engineering proposal without deciding the broader SAN theory. Conversely, agreement with a target neural metric does not prove a conscious percept.

10. Next experimental ladder

The appropriate next steps are ordered by increasing biological and ethical burden.

Stage A: frozen synthetic benchmark - completed for Draft 2

The frozen 2,520-run adversarial test is complete. It covers nonlinear dynamics, rank-deficient observations, correlated noise, delay, stimulation artifact, and targets outside the reachable subspace. It does not yet cover switching or nonminimum-phase plants. Those remain optional extensions, not reasons to discard the current adverse results.

Stage B: retrospective public data - completed on development sessions

The 14-session Allen component test establishes held-block scene reading and scene-conditioned target estimation in the declared development data. It preserves the five final sessions as a sealed future test and makes no writing claim.

Stage C: nonhuman or non-neural model system

Test read-model-write logic where complete state measurement and intervention are more defensible. Compare residual, feedforward, and standard feedback controllers with blinded analysis.

Stage D: regulated human research

Human stimulation would require an independently justified clinical or scientific purpose, device-specific dosimetry, ethics and regulatory review, adverse-event stopping rules, sham and blinding procedures, privacy protection, and a threat model for unauthorized neural writing. This paper supplies none of those approvals.

11. Ethical and interpretive boundary

A bidirectional perceptual interface raises risks beyond ordinary decoding. Targets can be wrong, coercive, or adversarial. A decoder can expose private information; a writer can alter perception or behavior. Participant-specific models are biometric and potentially health-sensitive data. Any future system must make authorization, revocation, auditability, and fail-safe shutdown part of the architecture rather than deployment policy added later.

The framework must also avoid a category error. A controller can move a measured state and improve a task without creating the intended qualia. Behavioral report, no-report measures, physiology, and neural readback provide different evidence and should not be collapsed into one consciousness score.

12. Conclusion

The 2017 Neural Lace proposal defined a concrete engineering problem before the later terminology existed: an artificial scene cannot safely be created by blindly adding a complete target pattern to an unknown ongoing pattern. The proposed solution was to estimate the current-to-target difference and transmit only that update. Alternating Transmission Protocol connected feedback and broadcast-like routing to the separate metabolic question carried by the ATP acronym. The 2022 SAN sources then expanded the operation into participant-specific phase differences, multimodal object rendering, PWD, and receive-transform-project architecture.

The modern formulation limits the residual to an observable and reachable projection, lets uncertainty and safety governors cancel the command, and requires comparison against simpler controllers and no control. Its negative synthetic results are part of the theory’s content: predictive residual control failed under measurement noise, was unsafe under observation-rank loss, and was worse than no control under combined stress. The new public-data test closes a narrower gap by showing that held-block scene reading and scene-conditioned target estimation work in 14 development sessions. It does not close the causal write-side gap. A successful codec still requires a frozen intervention test showing reachable state change, artifact rejection, perceptual or behavioral effect, safety, and advantage over matched alternatives.

Data, code, and reproducibility

The release package contains both synthetic benchmark generations, the frozen adversarial configuration and 2,520-run result, the public-data development configuration and 42-fold result, the hash-bound natural-scene stimulus metadata cache, deterministic runners, audits, and the validation report [12]. The five declared final-test session identities are recorded but were not opened.

Authorship and development history

The core ideas, scientific theories, hypotheses, and source program in this paper are Micah Blumberg’s original work developed across 2011-2026, with the specific brain-network protocol and differential scene-writing proposal publicly recorded in 2017 and the PWD / NAPOT extension documented in 2022. This research program was developed primarily before ChatGPT’s public release in November 2022; AI systems did not originate the work. AI tools have provided limited assistance with mathematical and coding formulations, literature retrieval, grammar and spelling checks, and medical, biological, and physical accuracy review under Blumberg’s direction. All scientific claims and release decisions remain the author’s responsibility.

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